

Blockchain-Based Supply Chain Financing & Verification Platform

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Project Overview

Blockchain

A decentralized platform for **supply chain financing**

Security

Secure document verification & transparent data sharing

Automation

Automated financing processes using **smart contracts**



Business Problems

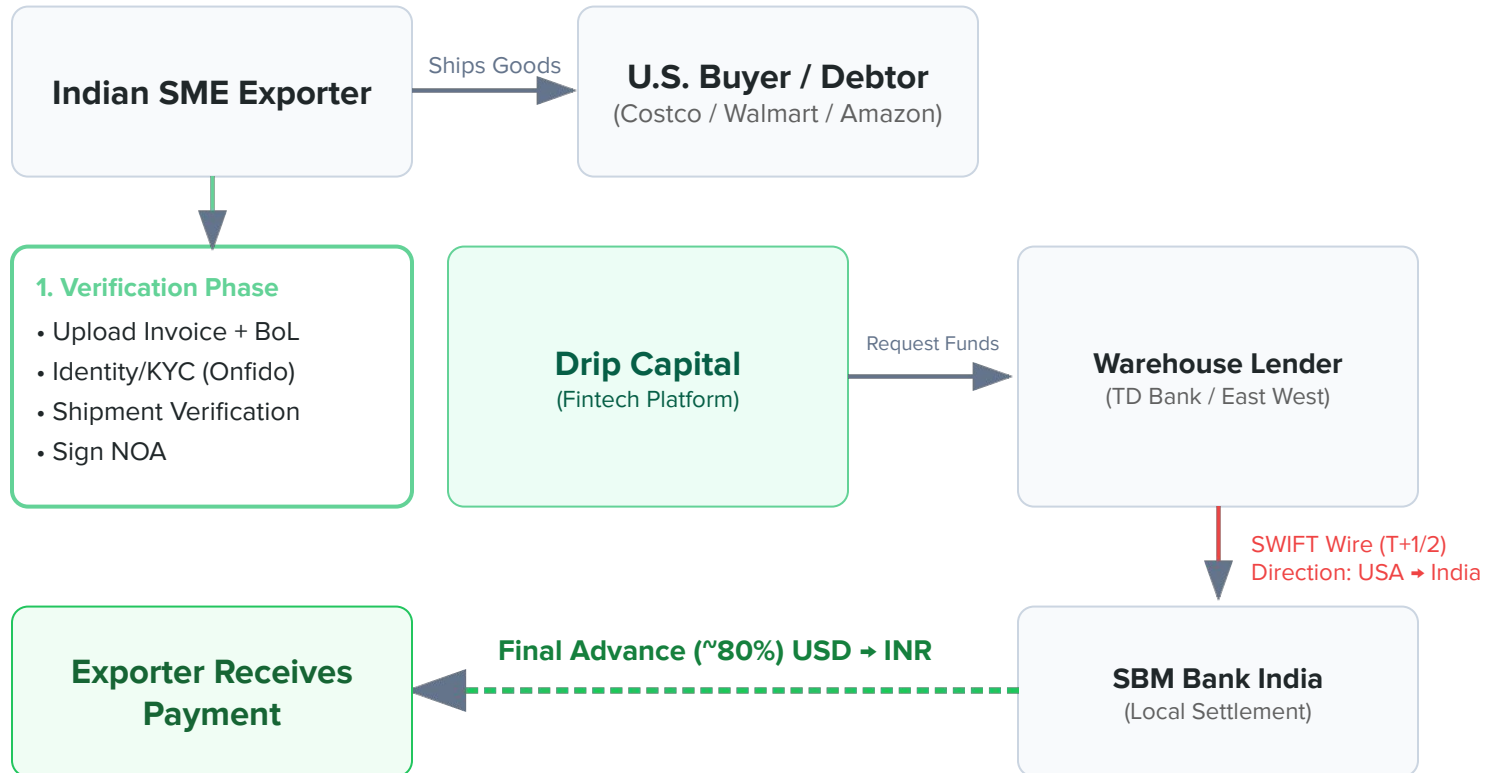
Key Challenges in Traditional Supply Chain Finance:

- Low transparency across parties
- Fragmented and siloed document storage
- **High risk of fraud (fake invoices, duplicate financing)**
- Slow and costly processes due to intermediaries
- Limited access to financing for SMEs

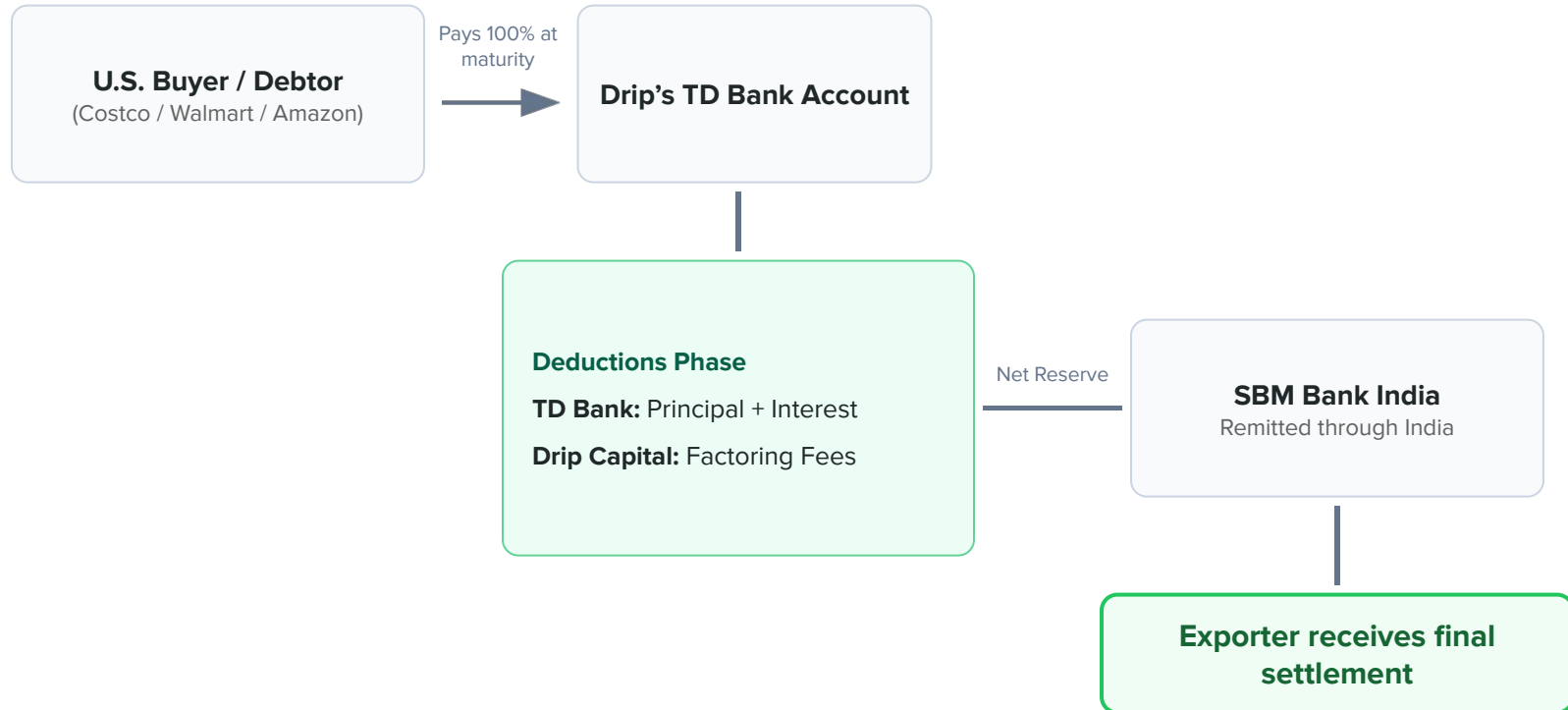
Lack of trust and inefficiency in the current system



Traditional Payment Flow for International Supply Chain Finance



Settlement Flow



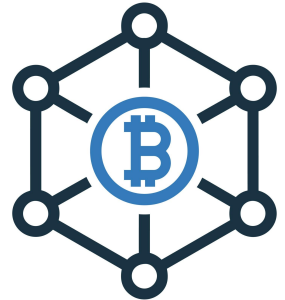
Proposed Solution

- Blockchain-based platform for trade finance
- Connects exporter, verifier, and lender
- Exporter registers trade documents
- Verifier approves document validity
- Lender funds verified invoices
- Smart contracts automate financing workflow
- Improves trust, efficiency, and transparency

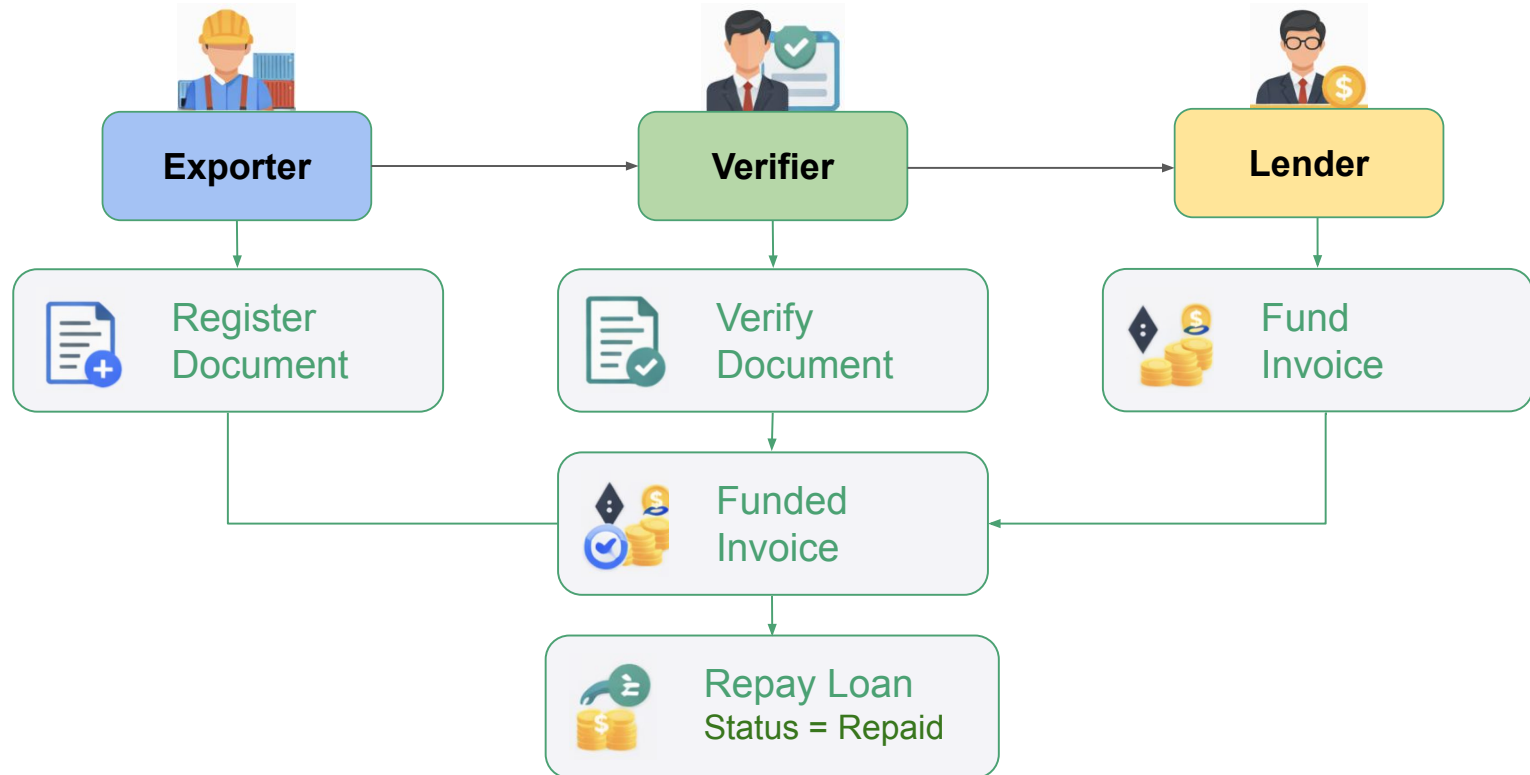


Why Blockchain

- Web2 depends on centralized control
- Centralized data creates trust issues
- Blockchain provides a shared ledger
- Records are tamper-resistant
- Smart contracts enforce rules automatically
- Better trust across multiple parties
- Higher transparency and accountability



Remix Smart Contract Workflow



Demo



Roles

Account 1: Owner / Verifier / Contract
Deployer:

**0x5B38Da6a701c568545dCfcB03F
cB875f56beddC4**

Account 2: Exporter:

**0xAb8483F64d9C6d1EcF9b849Ae
677dD3315835cb2**

Account 3: Lender:

**0x4B20993Bc481177ec7E8f571ceC
aE8A9e22C02db**



Workflow

1. Exporter registers a document
 2. Verifier verifies the document
 3. Lender funds the invoice
 4. Exporter repays the loan
- * Exporter may cancel unverified



Status Codes

0

Registered

1

Verified

2

Funded

3

Repaid

4

Cancelled

Technical Framework

→ Frontend

Web interface for exporters, lenders, and verifiers to interact with the platform - register documents, fund invoices, and track status in real time.

→ Smart Contract

Core business logic deployed on-chain. Manages the full invoice lifecycle across 5 states with role-restricted functions and immutable event logging.

→ Blockchain (Ethereum)

Decentralized ledger providing immutability, transparency, and trustless execution. Every state change is permanently recorded and publicly auditable.

Values & Future Directions



Fraud Prevention

Immutable on-chain document registration eliminates fake invoices and duplicate financing risks through cryptographic proof.



Efficiency & Speed

Smart contracts automate verification and fund release, replacing slow manual intermediaries and cutting processing time dramatically.



Full Transparency

Every state transition (Register → Verify → Fund → Repay) is publicly auditable, giving all parties real-time visibility.



SME Empowerment

Exporters gain timely access to working capital with lower costs, while lenders benefit from reliable, verified data.



Multi-chain deployment for broader ecosystem integration



Mobile DApp for SME exporters in emerging markets

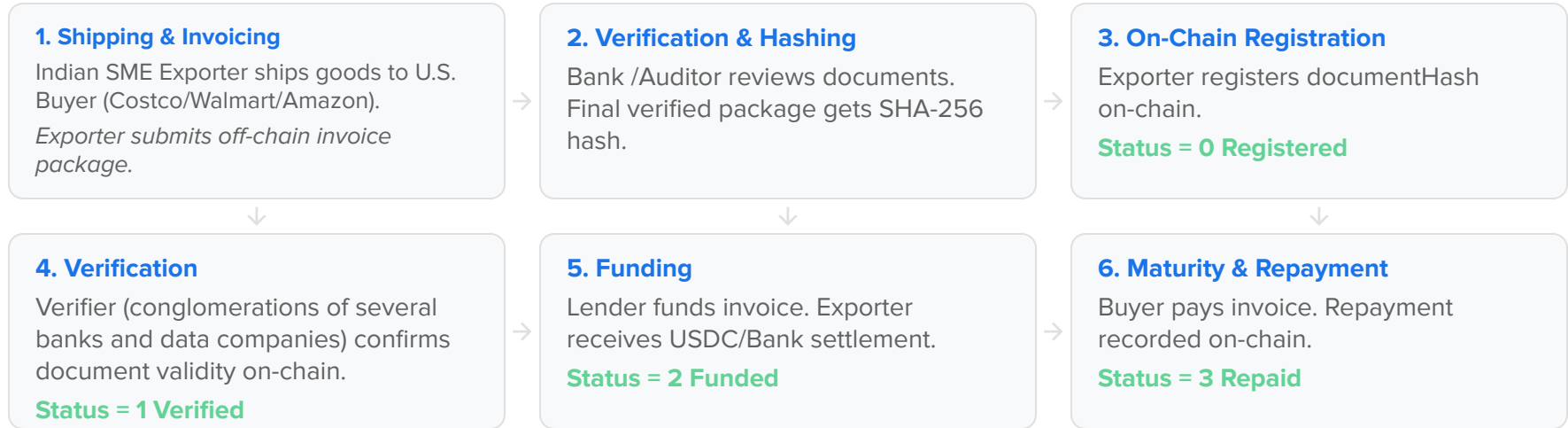


AI-driven document authenticity verification layer



Integration with trade finance APIs & banking networks

Final Payment Flow



Our Team



Yijun Cheng

Problem & Market

Identified core trade finance challenges and analyzed the SME financing gap across fragmented, centralized supply chain systems.



Shujia Zhang

Solution & Blockchain Rationale

Architected the proposed decentralized solution and articulated why blockchain uniquely solves transparency, trust, and efficiency gaps.



Wenyang Zu

Smart Contract Workflow & Demo

Designed and implemented the 5-state contract lifecycle in Solidity and led the live Remix demonstration across three accounts.



Jiayu Shi

Technical Framework & Values

Mapped the technical architecture and articulated the platform's core value propositions and future development roadmap.

Thank you!